

John Mwaura

📍 Nairobi, Kenya ✉ mwaura.ke.john@gmail.com ☎ +254705062319 🌐 in/john-mwaura-dev

SUMMARY

Systems engineer specialising in end-to-end product architecture, transactional payment infrastructure, and enterprise automation. Delivers production-grade systems across the full stack – from relational database schema design and API architecture to Vercel Edge deployments with a proven track record in M-Pesa Daraja integration, multi-tenant SaaS design, RBAC/RLS compliance, DNS security hardening, and ODPC-aligned data governance for African enterprise and corporate environments.

EXPERIENCE

Independent Systems Engineer & Solutions Architect

Free-Lancer

2023 – Present, Nairobi, Kenya

- **UWAZI – National Public Finance Transparency Platform**
- **Tech stack:** React, TypeScript, Node.js, PostgreSQL, Supabase, Vercel
- **Architecture:** Built a national-scale financial data engine with <100ms full-text search across normalized relational tables
- **Security:** Designed multi-tier RBAC + Row-Level Security (RLS) enforcing distinct data visibility for public, investigator, and administrator roles
- **Performance:** Deployed edge-optimized frontend on Vercel with GitHub Actions CI/CD, Lighthouse 99+ across all categories
- **IP registration:** Formally registered platform IP with Kenya Intellectual Property Institute (KIPI), securing ownership of system architecture and data schema
- **Niks Digital Connections – Enterprise E-Commerce & Payment Portal**
- **Tech stack:** Next.js 14, TypeScript, Supabase, M-Pesa Daraja, Flutterwave, Cloudinary, Vercel
- **Payments:** Built asynchronous payment pipeline integrating M-Pesa STK Push and Flutterwave multi-currency API; handles concurrent transaction queues, automated ledger reconciliation, and real-time callback status
- **Performance:** Automated Cloudinary image compression + Next.js 14 optimization → 99+ Lighthouse, sub-800ms LCP across all device profiles
- **Admin & logistics:** Full RBAC-gated inventory management, order lifecycle tracking, operational reporting, and delivery infrastructure covering all 47 Kenyan counties
- **WA-BOT-KE – Multi-Tenant B2B Workflow Automation Platform**
- **Tech stack:** Node.js, PostgreSQL, WhatsApp Cloud API, Render
- **Multi-tenancy:**
- **Automation engine:** Configuration-driven enabling non-technical operators to deploy custom conversation flows, escalation rules, and CRM integrations without code → reduced client manual support overhead by more than 80%
- **Karen West Natural Spring – Digital Identity & Search Infrastructure**
- **SEO & visibility:** Deployed structured data (LocalBusiness, Product, FAQPage) + Google Search Console manual indexing, dominant regional search category rankings
- **Email security:** Provisioned full email security stack – SPF, DKIM, DMARC, MX DNS records – securing domain reputation and maximizing transactional deliverability via Resend
- **Enterprise Communication Infrastructure (Retained Offering)**
- **Email & collaboration:** Provisions Google Workspace and Microsoft 365 environments with advanced DNS security (SPF, DKIM, DMARC, MX, subdomain isolation) to prevent domain spoofing and maximize corporate inbox deliverability
- **Data compliance:** Delivers ODPC-aligned data handling frameworks for Kenyan corporate clients, covering data classification policies, access logging, and breach notification readiness

EDUCATION

Advanced Systems Engineering Curriculum

Self-taught • 2023

- Full-stack architecture, security, and infrastructure engineering – applied across all production deployments.
- ODPC Data Privacy & Kenyan Compliance Ongoing
- Independent research applied to client data classification,.

SKILLS

Frontend Architecture: React 18, Next.js 14 (App Router), Vite, Framer Motion, TypeScript, Core Web Vitals Optimisation

Backend & Data Systems: Node.js, Express, PostgreSQL, tRPC, REST APIs, Supabase, Tailwind CSS, Prisma ORM, Redis

Payment Rails & Integrations: M-Pesa Daraja (Async STK Push), Stripe, WhatsApp Cloud API, Flutterwave (Multi-currency), OpenAI API, Resend

DevOps & Cloud Infrastructure: Docker, GitHub Actions, CI/CD, Vercel Edge, NGINX, Linux, Cloudinary CDN

Security & Enterprise Compliance: JWT Authentication, RBAC, Row-Level Security (RLS), Zod Schema Validation, SPF / DKIM / DMARC, ODPC Data Privacy Compliance
